

WELCOME ISLAND, ON, Canada

WMO: 717510

Lat: 48.369N

Lon: 89.119W

Elev: 694

StdP: 14.33

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.8	-11.4	-24.0	1.5	-15.1	-19.1	2.0	-10.4	36.2	6.6	33.3	6.5	15.8	300	0.757

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.9	75.4	64.7	72.5	63.5	70.0	62.6	67.4	71.8	65.5	69.9	63.9	68.2	11.9	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.8	97.5	69.8	63.8	90.8	67.7	62.1	85.4	66.4	32.3	71.9	30.7	70.2	29.6	68.2	78.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 33.0	29.0	25.6	-19.7	81.0	5.5	3.4	-23.7	83.4	-26.9	85.5	-30.0	87.4	-34.0	89.9		
(5)			-20.5	70.6	5.6	3.5	-24.6	73.2	-27.8	75.2	-31.0	77.2	-35.0	79.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	38.6	10.1	12.2	23.5	35.9	45.9	56.2	62.7	63.0	56.4	45.0	31.8	19.5
	DBStd	20.18	12.77	11.31	10.51	6.74	5.45	4.89	4.59	4.53	5.87	6.85	9.51	11.61
	HDD50	5393	1238	1059	822	425	151	7	0	0	14	185	546	946
	HDD65	9687	1703	1479	1287	874	592	268	102	94	262	619	996	1411
	CDD50	1250	0	0	0	1	24	192	393	403	205	32	0	0
	CDD65	69	0	0	0	0	0	3	30	32	4	0	0	0
	CDH74	148	0	0	0	0	0	15	85	43	5	0	0	0
	CDH80	11	0	0	0	0	0	1	8	2	0	0	0	0
Wind	WSAvg	12.3	14.4	12.9	12.1	12.4	11.1	9.8	10.1	10.3	11.2	13.7	14.5	15.0
Precipitation	PrecAvg	27.9	1.5	1.0	1.3	2.3	2.9	3.3	3.4	2.7	2.8	2.8	2.2	1.6
	PrecMax	37.0	3.0	2.5	2.3	7.9	7.2	7.3	8.3	5.2	5.1	6.0	5.9	3.4
	PrecMin	20.6	0.4	0.3	0.6	0.3	1.5	0.7	1.2	0.9	1.0	1.0	0.5	0.7
	PrecStd	3.8	0.6	0.5	0.5	1.4	1.3	1.6	1.6	1.1	1.1	1.3	1.3	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.7	38.8	47.6	55.8	67.4	76.0	80.7	78.5	73.6	64.2	53.2	41.2
		MCWB	34.9	36.1	41.6	46.2	54.7	63.2	66.4	66.4	62.5	55.2	48.2	38.9
	2%	DB	34.7	35.1	41.4	50.4	61.1	70.4	76.1	74.7	68.9	59.9	48.2	39.0
		MCWB	33.2	33.2	36.8	42.1	51.5	60.3	64.8	65.3	61.8	54.7	44.9	37.3
	5%	DB	32.3	32.2	38.2	47.1	57.2	66.6	72.9	72.1	66.4	57.3	46.1	36.9
		MCWB	31.2	30.6	34.6	40.9	49.7	58.5	63.7	63.9	61.4	52.8	43.5	35.5
	10%	DB	29.1	28.7	36.0	44.4	54.2	63.9	70.2	69.8	64.2	54.6	44.1	34.7
		MCWB	27.6	27.1	33.2	38.9	48.4	57.0	63.0	63.2	60.0	50.9	41.5	33.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.6	36.5	41.9	48.2	56.6	66.0	70.4	70.9	66.2	58.5	49.3	39.8
		MCDB	36.8	38.1	46.9	54.1	62.2	72.1	75.1	75.2	69.3	60.9	52.3	40.7
	2%	WB	33.3	33.6	37.4	44.0	53.6	62.0	67.1	68.0	63.9	55.6	46.0	37.5
		MCDB	34.5	34.9	40.2	48.4	58.1	67.3	72.3	71.8	66.7	58.9	47.8	38.6
	5%	WB	31.1	30.7	35.4	41.5	51.2	59.9	65.3	65.9	62.2	53.4	43.8	35.7
		MCDB	32.3	32.1	37.7	46.2	56.1	65.4	70.7	69.9	65.0	56.3	45.7	36.8
	10%	WB	27.5	27.1	33.7	39.3	48.9	58.0	63.7	64.3	60.6	51.4	41.9	33.5
		MCDB	29.0	28.5	35.8	43.8	53.5	62.6	69.0	68.0	63.3	54.3	43.8	34.7
Mean Daily Temperature Range	5% DB	MDBR	13.8	15.6	14.8	11.3	11.9	12.5	12.2	10.9	9.9	8.4	8.9	10.8
		MCDBR	11.3	14.1	13.2	15.0	16.7	16.9	16.0	13.7	12.1	11.5	9.1	9.4
		MCWBR	10.8	12.9	10.0	9.2	9.7	9.7	8.0	7.4	7.6	8.1	7.7	9.0
	5% WB	MCDBR	11.8	13.7	12.4	14.4	15.2	15.6	14.7	11.7	10.3	10.7	8.7	9.6
		MCWBR	11.4	12.7	9.8	9.7	10.3	10.4	8.8	7.4	7.6	8.5	7.8	9.4
Clear-Sky Solar Irradiance	taub	0.264	0.272	0.304	0.343	0.361	0.380	0.397	0.377	0.356	0.316	0.290	0.265	
	taud	2.415	2.389	2.387	2.375	2.410	2.383	2.383	2.438	2.483	2.561	2.530	2.462	
	Ebn at Noon	261	285	291	288	286	280	272	274	270	266	250	246	
	Edh at Noon	20	26	31	35	36	37	36	33	29	23	19	17	
All-Sky Solar Radiation	RadAvg	342	665	1075	1498	1736	1867	1881	1618	1160	685	362	279	
	RadStd	41	70	93	153	115	129	104	121	81	86	36	27	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (15 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page